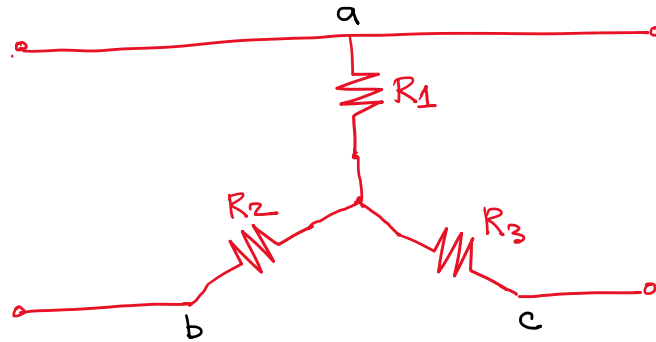


Delta-Y & Y-Delta steps

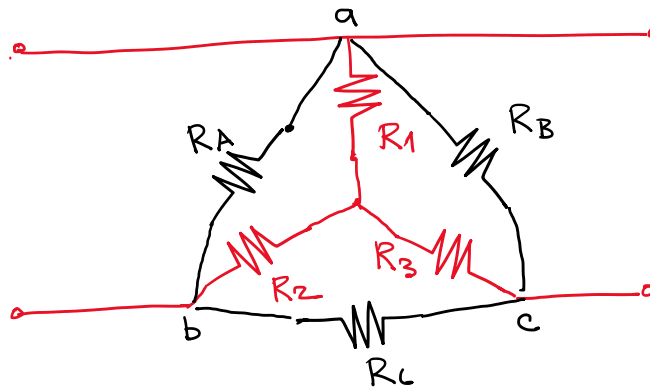
Sunday, December 21, 2025 8:16 PM



Steps to Convert a Y Network to Δ Network

□ label each node.

□ Draw the intended Δ -Network around the Y- Network.

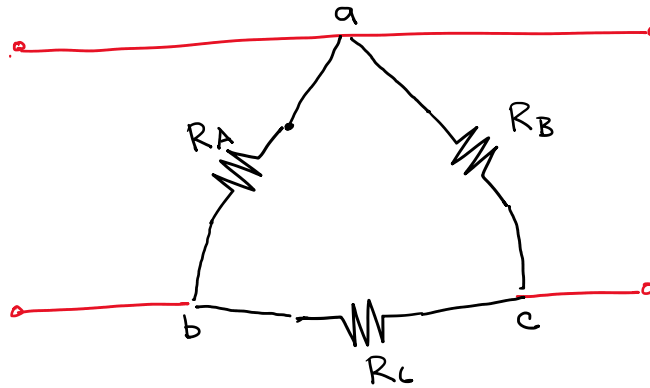


$$R_A = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_3} \quad \left[\begin{array}{l} \text{opposite Resistor of} \\ R_A \rightarrow R_3 \end{array} \right]$$

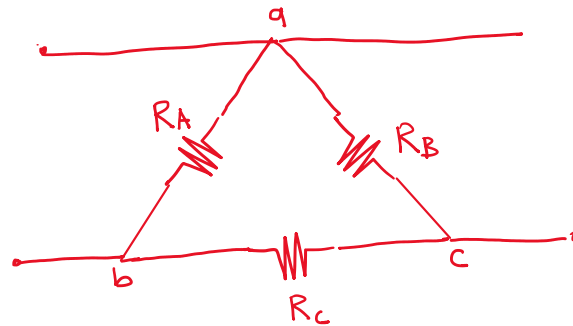
$$R_B = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_2} \quad \left[\begin{array}{l} \text{opposite Resistor of} \\ R_B \rightarrow R_2 \end{array} \right]$$

$$R_C = \frac{R_1 R_2 + R_2 R_3 + R_3 R_1}{R_1} \quad \left[\begin{array}{l} \text{opposite Resistor of} \\ R_C \rightarrow R_1 \end{array} \right]$$

□ Remove the Y Network, You are left with the Δ Network among the same 3 Nodes (a, b, c)

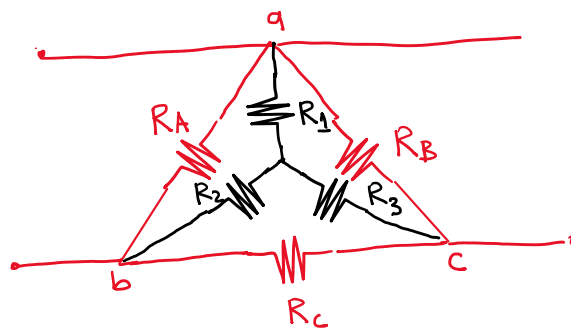


Steps to convert $\Delta \rightarrow Y$



□ label each node.

□ draw the intended Y-Network inside the Δ Network be



$$R_1 = \frac{R_A R_B}{R_A + R_B + R_C} \quad [R_1 \text{ is between } R_A \& R_B]$$

$$R_2 = \frac{R_A R_C}{R_A + R_B + R_C} \quad [R_2 \text{ is bet}^n R_A \& R_C]$$

$$R_3 = \frac{R_B R_C}{R_A + R_B + R_C} \quad [R_3 \text{ is bet}^n R_C \& R_B]$$

Finally Remove the Δ Network to obtain the
Corresponding Y Network

